

# Module 3: Token Standards (SPL Tokens)

## Solana Program Library Tokens

# SPL vs ERC Standards

## Solana Token Standards

- **SPL (Solana Program Library)**: Native token standard
- **SPL-Token**: Fungible tokens (like ERC-20)
- **SPL-Token-2022**: Enhanced token standard with extensions

## EVM Comparison

- **ERC-20**: Basic fungible tokens
- **ERC-721**: Non-fungible tokens
- **ERC-1155**: Multi-token standard

# Token Account Model

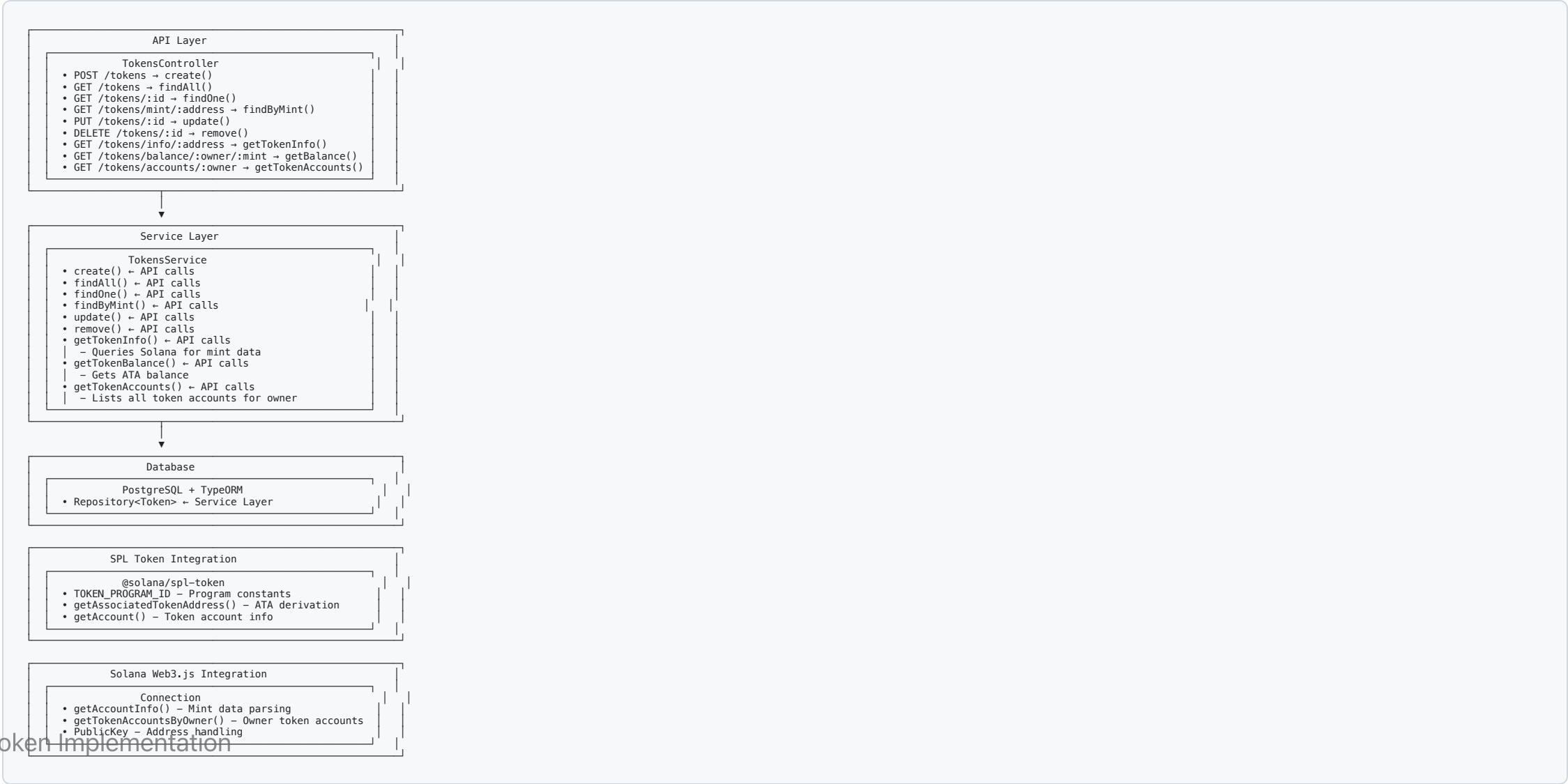
## Three Account Types

1. **Mint Account:** Token definition (supply, decimals, authorities)
2. **Token Account:** User's token balance
3. **Associated Token Account (ATA):** Deterministic token account address

## Key Differences from EVM

- **Separate Accounts:** Each token holding requires a token account
- **ATAs:** Automatic account derivation per wallet-token pair
- **No Direct Balance:** Balance stored in separate token accounts

# Architecture Overview



# Token Operations

## Creating a Token

```
// Create mint account
const mint = await createMint(
  connection,
  payer,
  mintAuthority,
  freezeAuthority,
  decimals
);

// Create token account
const tokenAccount = await getOrCreateAssociatedTokenAccount(
  connection,
  payer,
  mint,
  owner
);
```

## Token Transfer

```
// Transfer tokens between accounts
await transfer(
  connection,
  payer,
  sourceTokenAccount,
  destinationTokenAccount,
  owner,
  amount
);
```

## Minting Tokens

```
// Mint new tokens to account
await mintTo(
  connection,
  payer,
  mint,
  destinationTokenAccount,
  mintAuthority,
  amount
);
```

# SPL Token Extensions

## Token-2022 Features

- **Transfer Fees:** Charge fees on transfers
- **Confidential Transfers:** Private token amounts
- **Non-transferable Tokens:** Soul-bound tokens
- **Interest-bearing Tokens:** Automatic yield
- **Permanent Delegate:** Always-authorized delegate

## Metadata Extension

- **Token Metadata:** Name, symbol, description, image
- **Creator Verification:** Verify content creators
- **Royalties:** Automatic royalty payments



# NFT Implementation

## SPL Token NFTs

- **Supply = 1:** Single token per mint
- **Decimals = 0:** No fractional NFTs
- **Unique Metadata:** Each NFT has unique attributes

## Metaplex Standard

- **Metadata Account:** Stores NFT metadata
- **Master Edition:** Controls supply and royalties
- **Creator Verification:** Verify NFT creators

# Common Patterns

## Token Balance Checking

```
// Get token balance
const tokenAccounts = await getTokenAccountsByOwner(
  connection,
  ownerPublicKey,
  { mint: mintPublicKey }
);

const balance = tokenAccounts.value[0]?.account.data.parsed.info.tokenAmount.uiAmount;
```

## Batch Token Operations

```
// Create multiple token transfers
const instructions = [];
for (const transfer of transfers) {
  const ix = createTransferInstruction(
    transfer.source,
    transfer.destination,
    transfer.owner,
    transfer.amount
  );
  instructions.push(ix);
}

// Execute all in one transaction
const transaction = new Transaction().add(...instructions);
```

# Performance Considerations

## Optimization Strategies

- **ATA Caching:** Cache associated token addresses
- **Batch Operations:** Combine multiple transfers
- **Lazy Loading:** Load token data on demand
- **Index Optimization:** Efficient database queries

## Scalability

- **Parallel Processing:** Handle multiple token operations
- **Connection Pooling:** Efficient RPC connections
- **Rate Limiting:** Prevent API abuse

# Security Best Practices

## Token Security

- **Authority Validation:** Verify mint/freeze authorities
- **Amount Validation:** Prevent overflow/underflow
- **Account Ownership:** Verify token account ownership
- **Freeze Protection:** Handle frozen accounts

## Smart Contract Risks

- **Reentrancy:** Prevent reentrant calls
- **Access Control:** Proper permission checks
- **Input Validation:** Sanitize all inputs
- **Audit Requirements:** Security audits for custom tokens

# Next Steps

## Module 3 Implementation Tasks

-  Basic token CRUD operations
-  Token balance queries
-  Token creation/minting
-  Token transfer operations
-  NFT support
-  Token metadata management

## Related Modules

- **Module 2:** Transactions & Instructions
- **Module 4:** Account Abstraction
- **Module 10:** CPIs (Cross-Program Invocations)

# Thank You!

## Questions?

**Ready to explore Account Abstraction?**

Next: Module 4 - Account Abstraction 